

ECSE-4040

Digital Electronics

Lecture 5

Prof. T. Paul Chow

Bldg. CII, Room 6111

Rensselaer Polytechnic Institute

Troy, NY 12180-3590

Tel. (518)276-2910

FAX (518)276-2990

e-mail: chowt@rpi.edu

<https://lms.rpi.edu>

Lecture Outline

- MOSFET Layout
- Parameter Extraction
- SPICE Models
 - Level 1, BSIM3
 - Process Variations, Temperature Effects
- CMOS Latchup
- CMOS SOI

Layout Basics

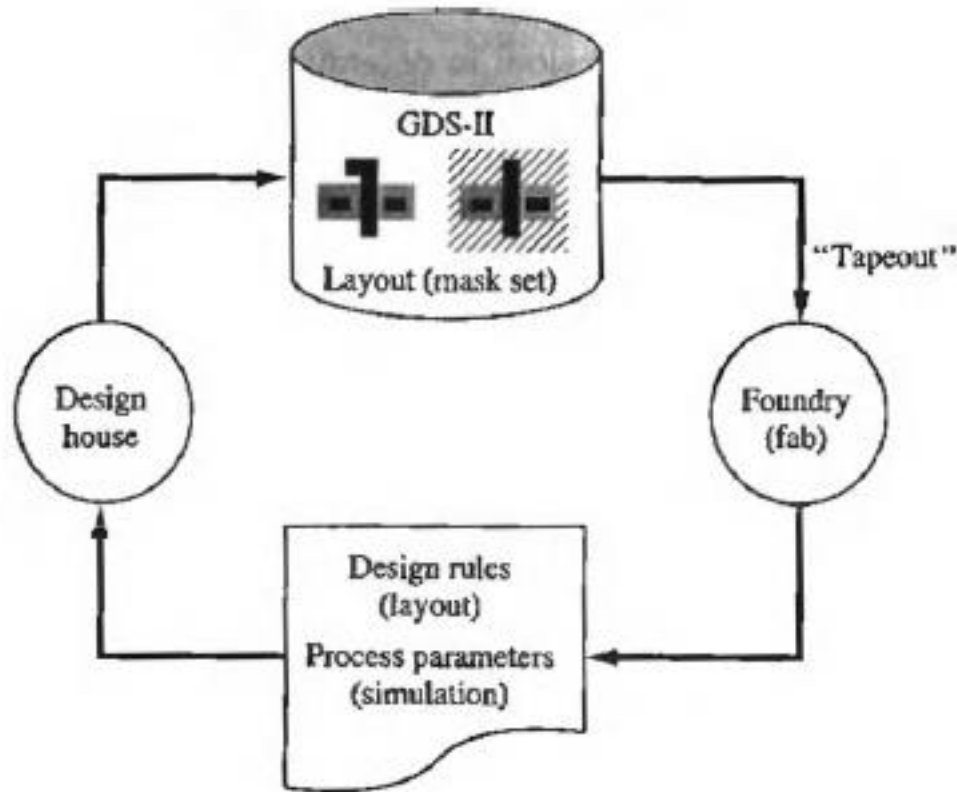
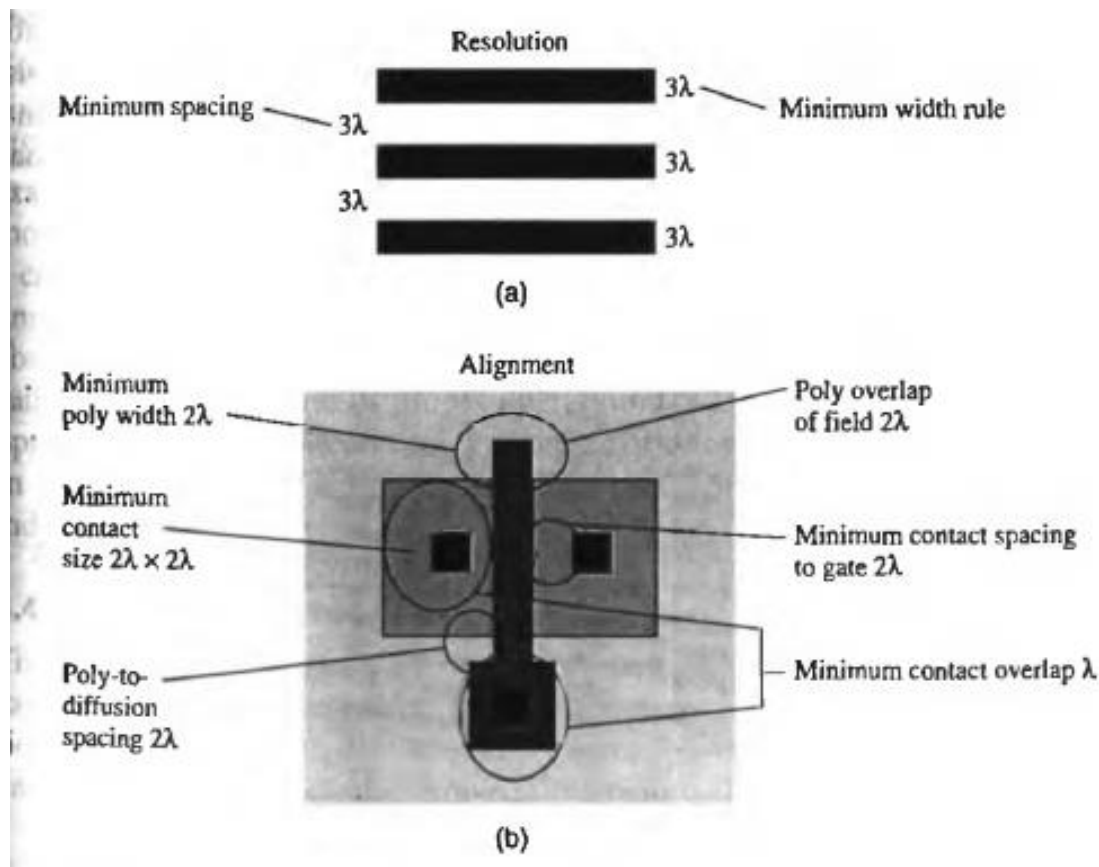


Figure 3.7

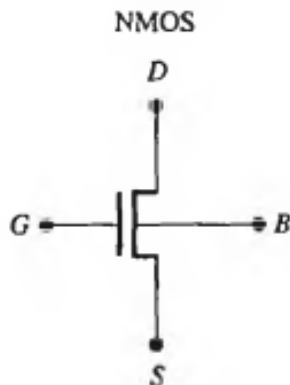
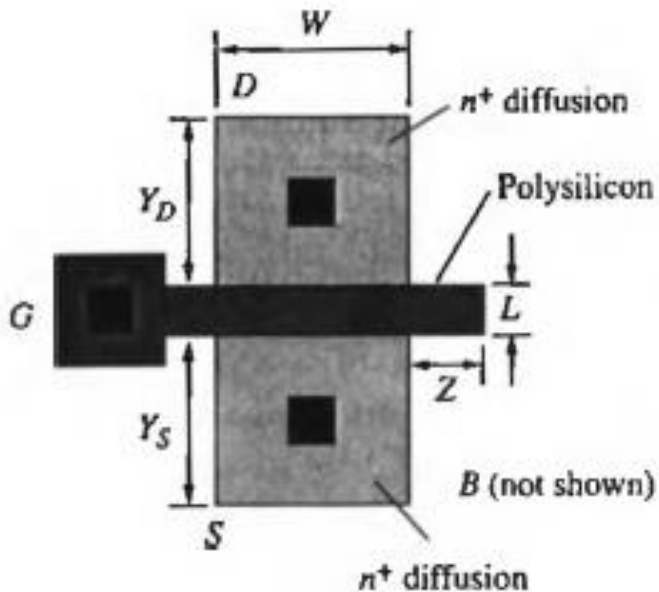
The flow of data between the design house and the foundry.

Design Rules

- Resolution and Alignment



MOSFET Layout and SPICE



MOSFET

Mxxx D G S B mname L = value W = value AD = value PD = value AS = value PS = value

Mxxx	The instance name of the MOSFET.
D	The drain node.
G	The gate node.
S	The source node.
B	The bulk node.
mname	The name of the model to be used. This is also where you distinguish between a PMOS and NMOS.
L	The length of the transistor.
W	The width of the transistor.
AD	The area of drain diffusion bottom region.
PD	The drain edges to be used for sidewall capacitance.
AS	The area of the source diffusion bottom region.
PS	The source edges to be used for sidewall capacitance.

MOSFET SPICE Model Level 1

- DC characteristics is defined by VTO, KP, LAMBDA, PHI, and GAMMA with process parameters, NSUB, TOX, UO, given

$$\text{PHI} = 2 \times \frac{kT}{q} \ln\left(\frac{\text{NSUB}}{n_i}\right) \qquad C_{ox} = \frac{\epsilon_{ox}}{\text{TOX}}$$

$$\text{GAMMA} = \frac{\sqrt{2e_{si}q \times \text{NSUB}}}{C_{ox}}$$

$$V_T = \text{VTO} + \text{GAMMA}(\sqrt{\text{PHI} - V_{BS}} - \sqrt{\text{PHI}})$$

MOSFET SPICE Model Level 1

- Linear Region

$$I_{DS} = \frac{W_{eff}}{L_{eff}} \frac{KP}{2} [2(V_{GS} - V_T)V_{DS} - V_{DS}^2](1 + LAMBDA \times V_{DS})$$

$$V_{GS} \geq V_T \quad V_{DS} \leq V_{GS} - V_T$$

- Saturation Region

$$I_{DS} = \frac{W_{eff}}{L_{eff}} \frac{KP}{2} (V_{GS} - V_T)^2 (1 + LAMBDA \times V_{DS})$$

$$V_{GS} \geq V_T \quad V_{DS} \geq V_{GS} - V_T$$

MOSFET SPICE Model Level 1

- Gate Capacitances

3 constant capacitors (CGSO, CGDO, CGBO) represent gate-source, gate-drain and gate-body capacitances respectively. $C = f(V)$, piecewise-linear, using Meyer Model.

- Junction Capacitances

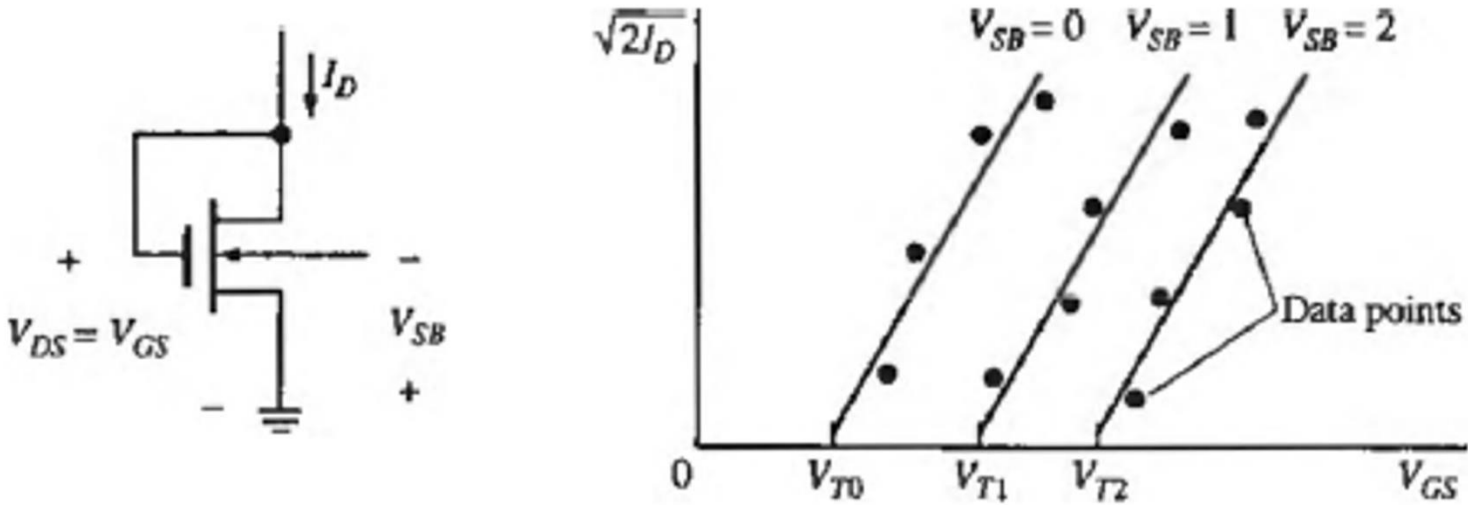
$$C_{JD} = \frac{CJ \times AD}{\left(1 - \frac{V_J}{PB}\right)^{MJ}} + \frac{CJSW \times PD}{\left(1 - \frac{V_J}{PB}\right)^{MJSW}} \quad \text{drain junction}$$

$$C_{JS} = \frac{CJ \times AS}{\left(1 - \frac{V_J}{PB}\right)^{MJ}} + \frac{CJSW \times PS}{\left(1 - \frac{V_J}{PB}\right)^{MJSW}} \quad \text{source junction}$$

- Resistances

RD, RS

Parameter Extraction

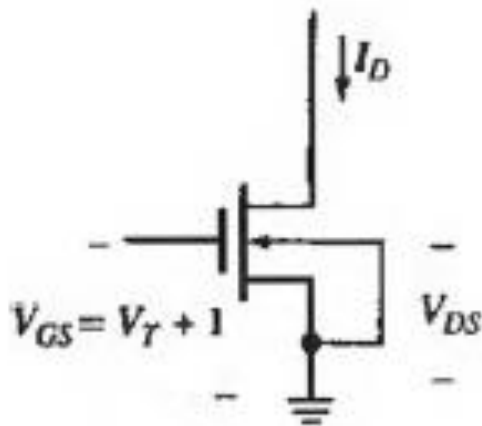


(1) V_{T0} from intercept

(2) $k = k' \frac{W}{L}$ from slope: $\sqrt{k} = \frac{\sqrt{2I_D}}{V_{GS} - V_T}$

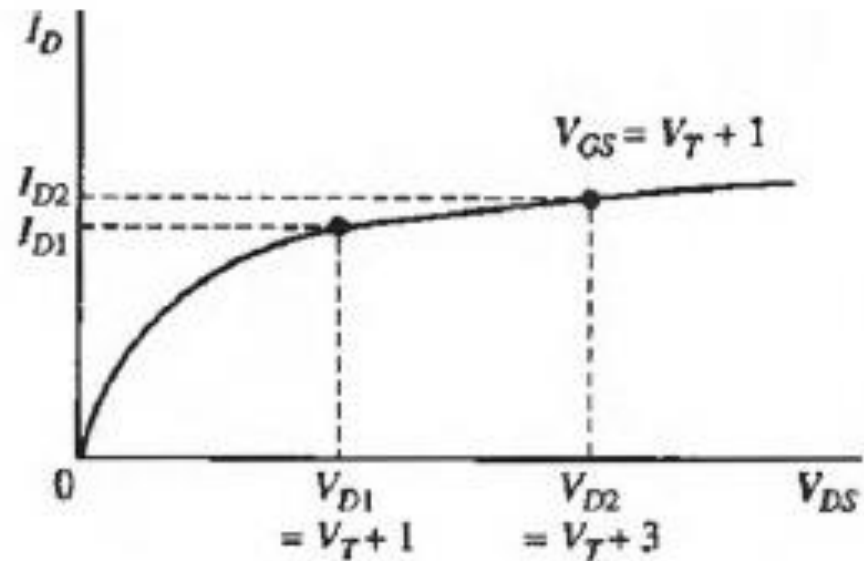
(3) $\gamma = \frac{V_T - V_{T0}}{\sqrt{V_{SB} - 2\phi_F} - \sqrt{2\phi_F}}$ $2\phi_F \cong -0.6 \text{ V}$, V_T is a function of V_{SB}

Parameter Extraction



(4) λ from:

$$\frac{I_{D2}}{I_{D1}} = \frac{1 + \lambda V_{D2}}{1 + \lambda V_{D1}}$$



MOSFET Level 1

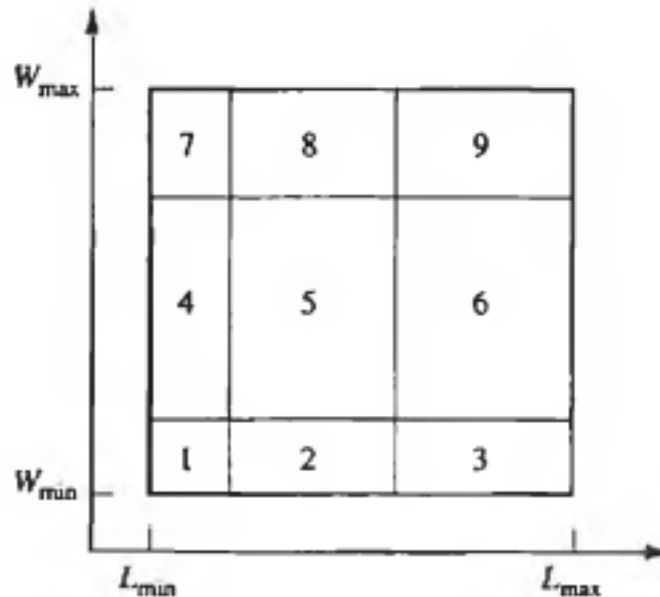
SPICE Parameters

Table 3.1
MOS LEVEL 1 parameters for SPICE

Symbol	Name	Parameter	Units	Default	Example	Symbol	Name	Parameter	Units	Default	Example
	LEVEL	Model Index		1			CJSW	Zero-bias bulk junction sidewall capacitance per meter of junction parameter	F/m	0.0	1.0E-9
V_{TO}	VTO	Zero-bias threshold voltage	V	0.0	1.0		mjsw	Bulk junction sidewall grading coefficient		0.33	
k'	KP	Transconductance parameter	A/V ²	2.0E-5	3.1E-5		JS	Bulk junction saturation current per square meter of junction area	A/m ²		1.0E-8
γ	GAMMA	Bulk threshold parameter	V ^{1/2}	0.0	0.37		t_{ox}	Oxide thickness	m	1.0E-7	1.0E-7
$2 \phi_F $	PHI	Surface potential	V	0.6	0.65		N_A or N_D	Substrate doping	cm ⁻³	0.0	4.0E15
λ	LAMBDA	Channel-length modulation	V ⁻¹	0.0	0.02		Q_{ss}/q	Surface state density	cm ⁻²	0.0	1.0E10
r_d	RD	Drain ohmic resistance	Ω	0.0	1.0		NFS	Fast surface state density	cm ⁻²	0.0	1.0E10
r_s	RS	Source ohmic resistance	Ω	0.0	1.0		TPG	Type of gate material: + 1 opposite to substrate - 1 same as substrate 0 Al gate		1.0	
ϕ_B	IS	Bulk junction saturation current	A	1.0E-14	1.0E-15		x_j	Metallurgical junction depth	m	0.0	1.0E-6
	PB	Bulk junction potential	V	0.8	0.87		L_D	Lateral diffusion	m	0.0	0.8E-6
	CGSO	Gate-source overlap capacitance per meter channel width	F/m	0.0	4.0E-11		μ_s	Surface mobility	cm ² /V-s	600	700
	CGDO	Gate-drain overlap capacitance per meter channel width	F/m	0.0	4.0E-11						
	CGBO	Gate-bulk overlap capacitance per meter channel length	F/m	0.0	2.0E-10						
	RSH	Drain and source diffusion sheet resistance	Ω /square	0.0	10.0						
C_{j0}	CJ	Zero-bias bulk junction bottom capacitance per square meter of junction area	F/m ²	0.0	2.0E-4						
m_j	MJ	Bulk junction bottom grading coefficient		0.5	0.5						

MOSFET SPICE Model BSIM3

- Binning Process
 - Need to find the device within the ranges of W , L value specified



MOSFET SPICE Model BSIM3

- Short Channel Threshold Voltage

$$V_T = V_{T0} + \gamma(\sqrt{2|\phi_F| + V_{SB}} - \sqrt{2|\phi_F|}) \quad \text{becomes}$$

$$V_T = V_{TH0} + \gamma(\sqrt{\text{PHI} - V_{BS}} - \sqrt{\text{PHI}})$$

$2|\phi_F|$ is called PHI and is given by

$$\text{PHI} = 2 \frac{kT}{q} \ln\left(\frac{N_{CH}}{n_i}\right)$$

Different doping for channel and substrate leads to

$$\gamma_1 = \frac{\sqrt{2\epsilon_{si}q \times N_{CH}}}{C_{ox}} \quad \gamma_2 = \frac{\sqrt{2\epsilon_{si}q \times N_{SUB}}}{C_{ox}}$$

MOSFET SPICE Model BSIM3

Oxide Capacitance $C_{ox} = \frac{\epsilon_{ox}}{TOX}$

, K1 and K2, are defined using γ_1 and γ_2 :

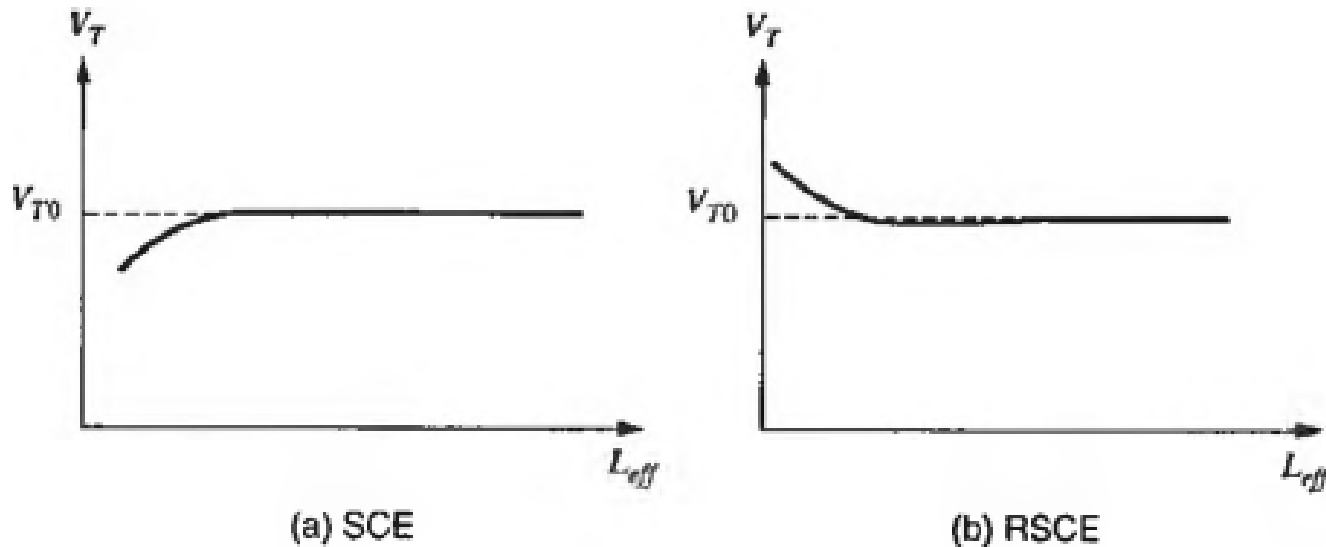
$$K1 = f(\gamma_2, PHI)$$

$$K2 = f(\gamma_1, \gamma_2, PHI)$$

K1, K2 model the vertical channel doping profile

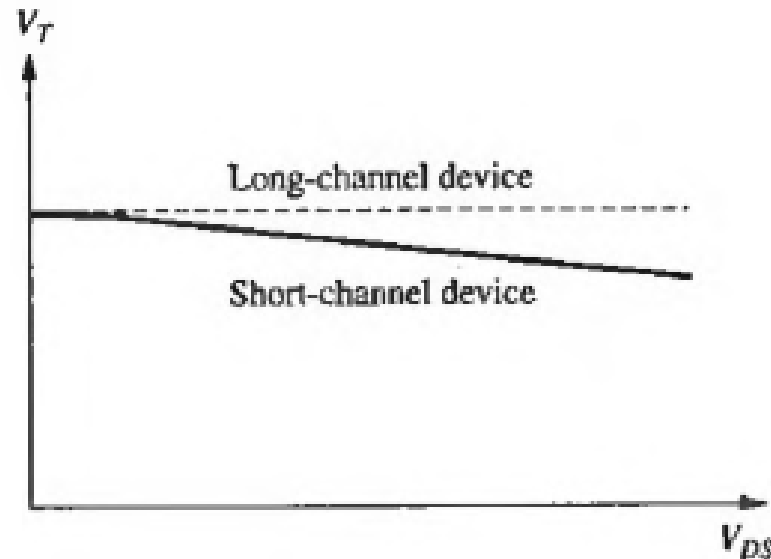
$$V_T = V_{TH0} + K1(\sqrt{PHI - V_{BS}} - \sqrt{PHI}) - K2 \times V_{BS}$$

MOSFET SPICE Model BSIM3



$$SCE = 0.64 \frac{\epsilon_{si}}{\epsilon_{ox}} \left(1 + \frac{x_j^2}{L_{el}^2} \right) \frac{t_{ox}}{L_{el}} \frac{W_{dm}}{L_{el}} V_{bi}$$

MOSFET SPICE Model BSIM3



$$DIBL = 0.80 \frac{\epsilon_{si}}{\epsilon_{ox}} \left(1 + \frac{x_j^2}{L_{el}^2} \right) \frac{t_{ox}}{L_{el}} \frac{W_{dm}}{L_{el}} V_{ds}$$

$$V_T = V_{TH0} + K1(\sqrt{\phi} - V_{BS} - \sqrt{\phi}) \\ - K2 \times V_{BS} - \eta V_{DS} - \Delta V_{SCE} + \Delta V_{NCE}$$

MOSFET SPICE Model BSIM3

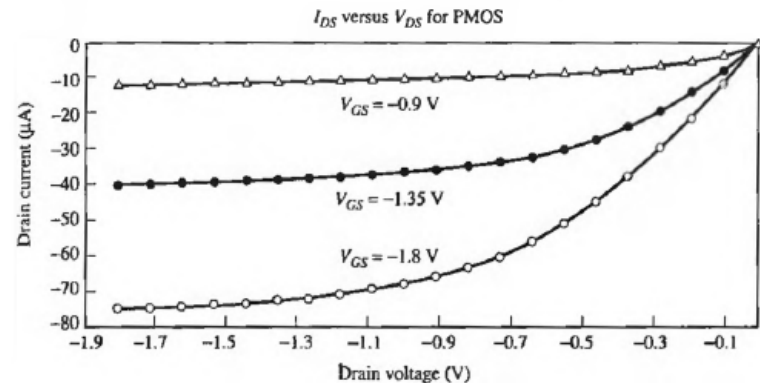
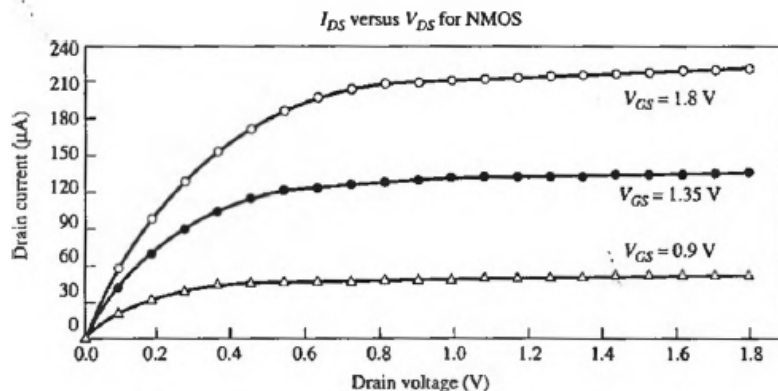
- Channel Mobility

UO modified by UA, UB, UC from vertical field

$$\mu_v = \frac{UO}{1 + (UA + UC + V_{BS}) \left(\frac{V_{GS} - V_T}{t_{ox}} \right) + UB \left(\frac{V_{GS} - V_T}{t_{ox}} \right)^2}$$

Use $E_c = \frac{2VSAT}{\mu_v}$ to guarantee v vs. E continuity

Ex. 3.4



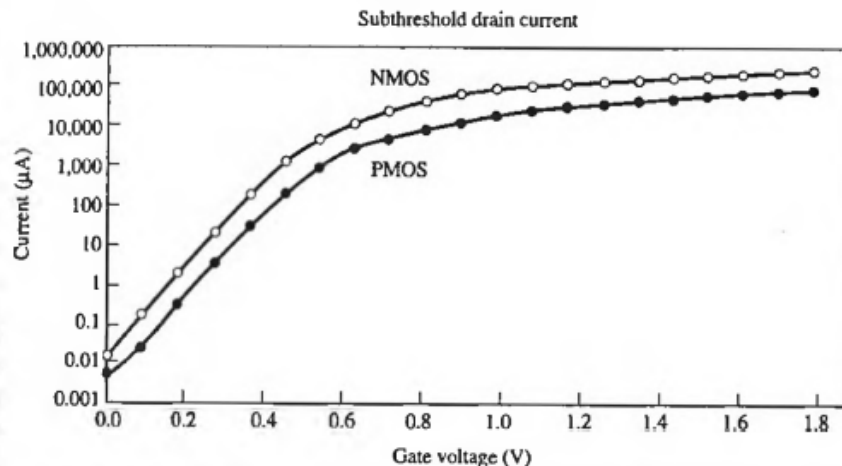
MOSFET SPICE Model BSIM3

- Subthreshold Current

$$I_{sub} = \beta_{sub} V_{th}^2 e^{(V_{GS} - V_T - V_{eff})/nV_n} (1 - e^{-V_{DS}/V_n})$$

$$\beta_{sub} = UO \times C_d W_{eff} / L_{eff} \quad C_d = \sqrt{\frac{q\epsilon_{si} NCH}{PHI}}$$

Ex. 3.6



MOSFET SPICE Model BSIM3

- Capacitance Models

Total charge is conserved, more accurate than Level 1

C_{gs} , C_{gd} , C_{ds} computed based on regions of operation

$$C_{JD} = \frac{CJ \times AD}{\left(1 - \frac{V_f}{PB}\right)^{MJ}} + \frac{CJSWG \times PD}{\left(1 - \frac{V_f}{PB}\right)^{MJSWG}} \quad \text{drain junction}$$

$$C_{JS} = \frac{CJ \times AS}{\left(1 - \frac{V_f}{PB}\right)^{MJ}} + \frac{CJSWG \times PS}{\left(1 - \frac{V_f}{PB}\right)^{MJSWG}} \quad \text{source junction}$$

- Source/Drain Resistance

R_S , R_D in Ohms

R_{SH} in Ohms per square

$$R_{DS} = R_D + R_S = \frac{R_{DSW}}{W_{eff}}$$

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Model Flags

Name	Units	Default	Comments
<i>VERSION</i>	-	3.11	Selects from BSIM3 Versions 3.0, 3.1, 3.11 (version=3.11 is the HSPICE97.4 equivalent to BSIM3v3 version 3.1)
<i>PARAMCHK</i>	-	0	PARAMCHK=1 will check some model parameters for range compliance
<i>BINFLAG</i>	-	0	Uses wref, lref when set > 0.9
<i>MOBMOD</i>	-	1	mobility model selector
<i>CAPMOD</i>	-	0	Selects from Charge models 0,1,2 (default deviates from BSIM3v3 = 2)
<i>NOIMOD</i>	-	1	Berkeley noise model flag
<i>NLEV</i>	-	0 (off)	Star-Hspice noise model flag (non-zero overrides NOIMOD) (Star-Hspice specific)
<i>NQSMOD</i>	-	0 (off)	NQS Model is not supported in Level 49
<i>SFVTFLAG</i>	-	1 (on)	Spline function for Vth (Star-Hspice specific)
<i>VFBFLAG</i>	-	0 (off)	UFB selector for CAPMOD=0 (Star-Hspice specific)

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Basic Model Parameters

Name	Units	Default	Comments
<i>TOX</i>	m	150e-10	gate oxide thickness
<i>XJ</i>	m	0.15e-6	junction depth
<i>NGATE</i>	cm ⁻³	infinite	Poly gate doping concentration
<i>VTH0</i> (<i>VTHO</i>)	V	0.7	threshold voltage of long channel device at $V_{bs} = 0$ and small V_{ds} (typically 0.7 for n-channel, - 0.7 for p-channel)
<i>NSUB</i>	cm ⁻³	6.0e16	substrate doping concentration
<i>NCH</i>	cm ⁻³ (see Note 6)	1.7e17	peak doping concentration near interface
<i>NLX</i>	m	1.74e-7	lateral nonuniform doping along channel
<i>K1</i>	V ^{1/2}	0.50	first-order body effect coefficient
<i>K2</i>	-	-0.0186	second-order body effect coefficient
<i>K3</i>	-	80.0	narrow width effect coefficient
<i>K3B</i>	1/V	0	body width coefficient of narrow width effect
<i>W0</i>	m	2.5e-6	narrow width effect coefficient

6. If n_{ch} is not given, and $GAMMA_I$ is given, n_{ch} is calculated from:

$$n_{ch} = \frac{GAMMA_I^2 C_{OX}^2}{2q\epsilon_{si}}$$

If both n_{ch} and $GAMMA_I$ are not given, n_{ch} defaults to 1.7e17 per cubic meter and $GAMMA_I$ is calculated from n_{ch} .

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Name	Units	Default	Comments
<i>DVT0W</i>	1/m	0	narrow width coefficient 0, for V_{th} , at small L
<i>DVT1W</i>	1/m	5.3e6	narrow width coefficient 1, for V_{th} , at small L
<i>DVT2W</i>	1/V	-0.032	narrow width coefficient 2, for V_{th} , at small L
<i>DVT0</i>	-	2.2	short channel effect coefficient 0, for V_{th}
<i>DVT1</i>	-	0.53	short channel effect coefficient 1, for V_{th}
<i>DVT2</i>	1/V	-0.032	short channel effect coefficient 2, for V_{th}
<i>ETA0</i>	-	0.08	subthreshold region DIBL (Drain Induced Barrier Lowering) coefficient
<i>ETAB</i>	1/V	-0.07	subthreshold region DIBL coefficient
<i>DSUB</i>	-	DROUT	DIBL coefficient exponent in subthreshold region
<i>VBM</i>	V	-3.0	maximum substrate bias, for V_{th} calculation
<i>U0</i>	$\text{cm}^2/\text{V}/\text{sec}$	670 nmos 250 pmos	low field mobility at $T = T_{REF} = T_{NOM}$
<i>UA</i>	m/V	2.25e-9	first-order mobility degradation coefficient
<i>UB</i>	m^2/V^2	5.87e-19	second-order mobility degradation coefficient
<i>UC</i>	1/V	-4.65e-11 or -0.0465	body bias sensitivity coefficient of mobility -4.65e-11 for $\text{MOBMOD}=1,2$ or, -0.0465 for $\text{MOBMOD} = 3$
<i>A0</i>	-	1.0	bulk charge effect coefficient for channel length

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Name	Units	Default	Comments
<i>AGS</i>	1/V	0.0	gate bias coefficient of A_{bulk}
<i>B0</i>	m	0.0	bulk charge effect coefficient for channel width
<i>B1</i>	m	0.0	bulk charge effect width offset
<i>KETA</i>	1/V	-0.047	body-bias coefficient of bulk charge effect
<i>VOFF</i>	V	-0.08	offset voltage in subthreshold region
<i>VSAT</i>	m/sec	8e4	saturation velocity of carrier at $T = T_{REF} = T_{NOM}$
<i>A1</i>	1/V	0	first nonsaturation factor
<i>A2</i>	-	1.0	second nonsaturation factor
<i>RDSW</i>	ohm · μm	0.0	parasitic source drain resistance per unit width
<i>PRWG</i>	1/V	0	gate bias effect coefficient of <i>RDSW</i>
<i>PRWB</i>	1/V ^{1/2}	0	body effect coefficient of <i>RDSW</i>
<i>WR</i>	-	1.0	width offset from W_{eff} for R_{ds} calculation
<i>NFACTOR</i>	-	1.0	subthreshold region swing
<i>CIT</i>	F/m ²	0.0	interface state capacitance
<i>CDSC</i>	F/m ²	2.4e-4	drain/source and channel coupling capacitance
<i>CDSCD</i>	F/Vm ²	0	drain bias sensitivity of <i>CDSC</i>
<i>CDSCB</i>	F/Vm ²	0	body coefficient for <i>CDSC</i>
<i>PCLM</i>	-	1.3	coefficient of channel length modulation values ≤ 0 will result in an error message and program exit.

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Name	Units	Default	Comments
<i>PDIBLC1</i>	-	0.39	DIBL (Drain Induced Barrier Lowering) effect coefficient 1
<i>PDIBLC2</i>	-	0.0086	DIBL effect coefficient 2
<i>PDIBLCB</i>	1/V	0	body effect coefficient of DIBL effect coefficients
<i>DROUT</i>	-	0.56	length dependence coefficient of the DIBL correction parameter in R_{out}
<i>PSCBE1</i>	V/m	4.24e8	substrate current induced body effect exponent 1
<i>PSCBE2</i>	V/m	1.0e-5	substrate current induced body effect coefficient 2
<i>PVAG</i>	-	0	gate dependence of Early voltage
<i>DELTA</i>	V	0.01	effective V_{ds} parameter
<i>ALPHA0</i>	m/V	0	the first parameter of impact ionization current
<i>BETA0</i>	V	30	the second parameter of impact ionization current
<i>RSH</i>	0.0	ohm/square	source/drain sheet resistance in ohm per square

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

AC and Capacitance Parameters

Name	Units	Default	Comments
<i>XPART</i>	-	1	charge partitioning rate flag (default deviates from BSIM3V3=0)
<i>CGSO</i>	F/m	p1 (see Note1)	non-LDD region source-gate overlap capacitance per unit channel length
<i>CGDO</i>	F/m	p2 (see Note2)	non-LDD region source-gate overlap capacitance per unit channel length
<i>CGBO</i>	F/m	0	gate-bulk overlap capacitance per unit channel length
<i>CGS1</i>	F/m	0.0	lightly doped source-gate overlap region capacitance
<i>CGD1</i>	F/m	0.0	lightly doped drain-gate overlap region capacitance
<i>CKAPPA</i>	F/m	0.6	coefficient for lightly doped region overlap capacitance fringing field capacitance
<i>CF</i>	F/m	(see Note3)	fringing field capacitance
<i>CLC</i>	m	0.1e-6	constant term for the short channel model
<i>CLE</i>	-	0.6	exponential term for the short channel model

1. If C_{gso} is not given, it is calculated as follows:
If (dlc is given and is greater than 0.0), then,
 $cgso = p1 = \max(0, dlc * cox - cgs1)$
Otherwise, $cgso = 0.6 * xj * cox$
2. If C_{gdo} is not given, it is calculated as follows:
if (dlc is given and is greater than 0.0), then,
 $cgdo = p2 = \max(0, dlc * cox - cgd1)$
Otherwise $cgdo = 0.6 * xj * cox$
3. If C_f is not given, it is calculated using:

$$C_f = \frac{2\epsilon_{ox}}{\pi} \log \left(1 + \frac{4 \times 10^{-7}}{T_{ox}} \right)$$

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Length and Width Parameters

Name	Units	Default	Comments
<i>WINT</i>	m	0.0	width offset fitting parameter from I-V without bias
<i>WLN</i>	-	1.0	power of length dependence of width offset
<i>WW</i>	m^{WWN}	0.0	coefficient of width dependence for width offset
<i>WWN</i>	-	1.0	power of width dependence of width offset.
<i>WWL</i>	$m^{WWN} * m^{WLN}$	0.0	coefficient of length and width cross term for width offset
<i>DWG</i>	m/V	0.0	coefficient of Weff's gate dependence
<i>DWB</i>	$m/V^{1/2}$	0.0	coefficient of Weff's substrate body bias dependence
<i>LINT</i>	m	0.0	length offset fitting parameter from I-V without bias
<i>LL</i>	m^{LLN}	0.0	coefficient of length dependence for length offset
<i>LLN</i>	-	1.0	power of length dependence of length offset
<i>LW</i>	m^{LWN}	0.0	coefficient of width dependence for length offset
<i>LWN</i>	-	1.0	power of width dependence of length offset
<i>LWL</i>	$m^{LWN} * m^{LLN}$	0.0	coefficient of length and width cross term for length offset
<i>DLC</i>	m	LINT	length offset fitting parameter from CV
<i>DWC</i>	m	WINT	width offset fitting parameter from CV

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Temperature Parameters

Name	Units	Default	Comments
<i>KT1</i>	V	0.0	temperature coefficient for Vth
<i>KT1L</i>	m-V	0.0	temperature coef. for channel length dependence of Vth
<i>KT2</i>	-	0.022	body bias coefficient of Vth temperature effect
<i>UTE</i>	-	-1.5	mobility temperature exponent
<i>UA1</i>	m/V	4.31e-9	temperature coefficient for UA
<i>UB1</i>	(m/V) ²	-7.61e-18	temperature coefficient for UB
<i>UC1</i>	m/V ²	-5.69e-11	temperature coefficient for UC
<i>AT</i>	m/sec	3.3e4	temperature coefficient for saturation velocity
<i>PRT</i>	ohm-um	0	temperature coefficient for RDSW
<i>XTI</i>	-	3.0	junction current temperature exponent

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Bin Description Parameters

Name	Units	Default	Comments
<i>LMIN</i>	m	0.0	maximum channel length
<i>LMAX</i>	m	1.0	maximum channel length
<i>WMIN</i>	m	0.0	minimum channel width
<i>WMAX</i>	m	1.0	maximum channel width
<i>BINUNIT</i>	-	0	assumes weff,leff,wref,lref units are in microns when BINUNIT=1 or meters otherwise

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Process Parameters

Name	Units	Default	Comments
$GAMMA_1$	$V^{1/2}$	see Note 8	body effect coefficient near the surface
$GAMMA_2$	$V^{1/2}$	see Note 9	body effect coefficient in the bulk
VBX	V	see Note 10	VBX at which the depletion region width equals XT
XT	m	1.55e-7	doping depth
VBI	V	see Note 11	drain and source junction built-in potential

8. If $GAMMA_1$ is not given, it is calculated using:

$$GAMMA_1 = \frac{\sqrt{2q\epsilon_{si}n_{ch}}}{C_{ox}}$$

9. If $GAMMA_2$ is not given, it is calculated using:

$$GAMMA_2 = \frac{\sqrt{2q\epsilon_{si}n_{sub}}}{C_{ox}}$$

10. If V_{bx} is not given, it is calculated using:

$$V_{bx} = \phi_s - \frac{qn_{ch}X_t^2}{2\epsilon_{si}}$$

11. If V_{bi} is not given, it is calculated using:

$$V_{bi} = \frac{k_B T}{q} \log\left(\frac{n_{ch} N_{DS}}{n_i^2}\right)$$

where N_{DS} is $1e20/cm^2$

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Noise Parameters

Name	Units	Default	Comments
<i>NIOA</i>	-	1.0e20 nmos 9.9e18 pmos	body effect coefficient near the surface
<i>NOIB</i>	-	5.0e4 nmos 2.4e3 pmos	body effect coefficient in the bulk
<i>NOIC</i>	-	-1.4e-12 nmos 1.4e-12 pmos	VBX at which the depletion region width equals XT
<i>EM</i>	V/m	4.1e ⁷	doping depth
<i>AF</i>	-	1.0	drain and source junction built-in potential
<i>KF</i>	-	1.0	flicker exponent
<i>EF</i>	-	0.0	flicker noise parameter

MOSFET BSIM3 SPICE Level 49

Default Model Parameters

Junction Parameters

Name	Units	Default	Comments
<i>ACM</i>	-	0	Area calculation method selector (Star-Hspice specific)
<i>JS</i>	A/m ²	0.0	bulk junction saturation current (Default deviates from BSIM3v3 = 1.0e ⁻⁴)
<i>JSW</i>	A/m	0.0	sidewall bulk junction saturation current
<i>NJ</i>	-	1	emission coefficient (not used with ACM=3)
<i>CJ</i>	F/m ²	5.79e ⁻⁴	zero-bias bulk junction capacitance (Default deviates from BSIM3v3 = 5.0e ⁻⁴)
<i>CJSW</i>	F/m	0.0	zero-bias sidewall bulk junction capacitance (Default deviates from BSIM3v3 = 5.0e ⁻¹⁰)
<i>CJSWG</i>	F/m	CJSW	zero-bias gate-edge sidewall bulk junction capacitance (not used with ACM=0-3)
<i>PB, PHIB</i>	V	1.0	bulk junction contact potential
<i>PBSW</i>	V	1.0	sidewall bulk junction contact potential
<i>PBSWG</i>	V	PBSW	gate-edge sidewall bulk junction contact potential (not used with ACM=0-3)
<i>MJ</i>	-	0.5	bulk junction grading coefficient
<i>MJSW</i>	-	0.33	sidewall bulk junction grading coefficient
<i>MJSWG</i>	-	MJSW	gate-edge sidewall bulk junction grading coefficient (not used with ACM=0-3)

MOSFET Additional Limitations

- Process Variations
- Temperature Effects
- Voltage Limitations
 - Voltage breakdown on Gate Oxide =>
Gate Protection Circuits
- CMOS Latchup (Loss of Gate Control)
 - SCR (or thyristor) triggering
 - Ohmic Contacts to well and substrate need to have low resistance values

Process Variations

- **Sources of Variations**

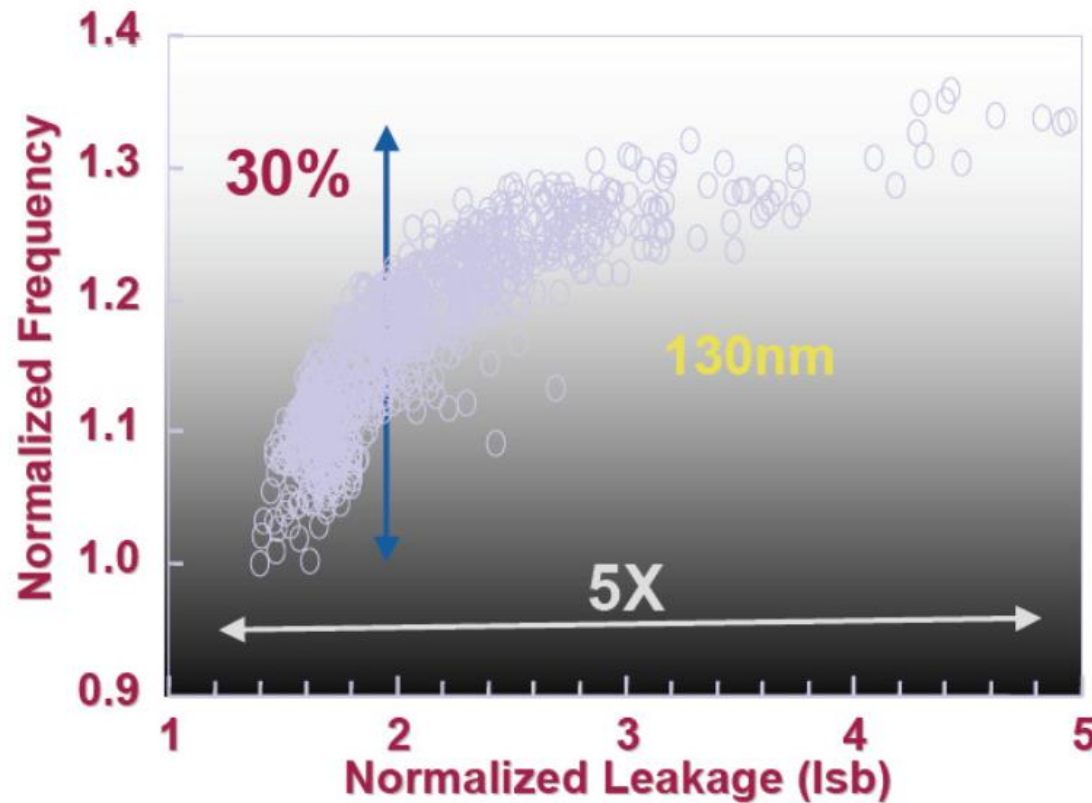
- Gate Oxide thickness (T_{OX})
- Random Doping Fluctuations (RDF)
- Device geometry, Lithography in nanometer region
- Transistor Threshold Voltage (V_T)
 - ◆ Sub threshold current, leakage, power, frequency

- **Range of Variations**

- 100% V_T variation across a modern chip
- 30% speed variation across a wafer
- 100% leakage (static power) variation in a wafer

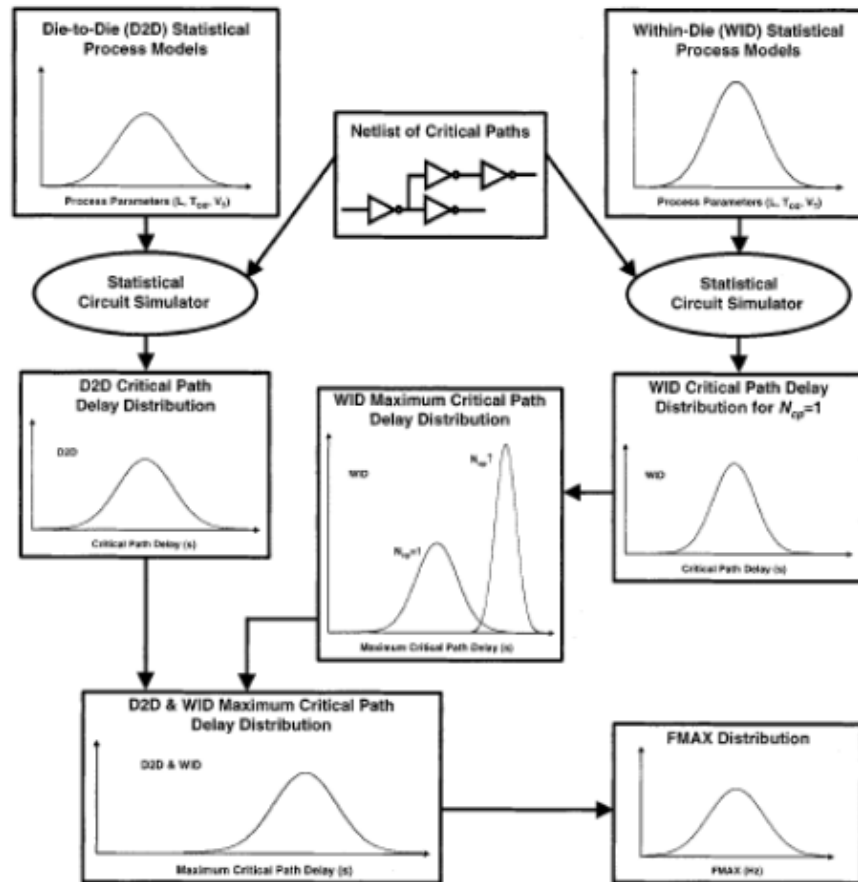
**J.H.Patel, 2008*

Process Variations



Source: Intel

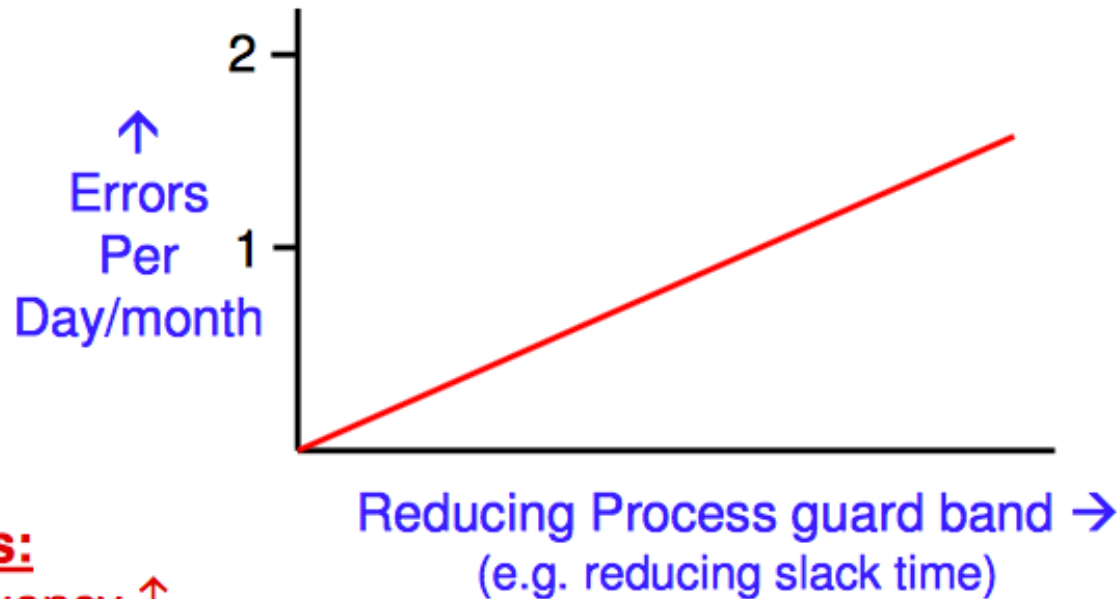
FAMX Statistical Analysis



***J.H.Patel, 2008**

Source: Bowman, K.A.; Duvall, S.G.; Meindl, J.D., "Impact of die-to-die and within-die parameter fluctuations on the maximum clock frequency distribution for gigascale integration," *Solid-State Circuits, IEEE Journal of*, vol.37, no.2, pp.183-190, Feb 2002

Errors and Process Variations



Parameters:

Clock Frequency ↑
Supply Voltage ↓
Ambient Temperature ↑
Gate and Pin Switching rate ↑
Years of Aging ↑
Process Variations ↑

Errors:

All are timing errors
No spontaneous bit flips

***J.H.Patel, 2008**

Temperature Effects

- Carrier Mobility

Inversion carrier mobility has different temperature dependence from bulk mobility.

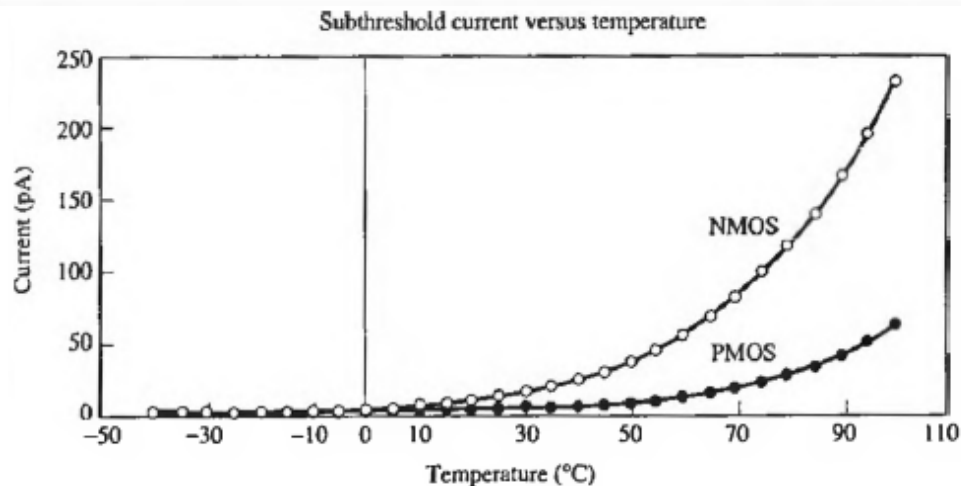
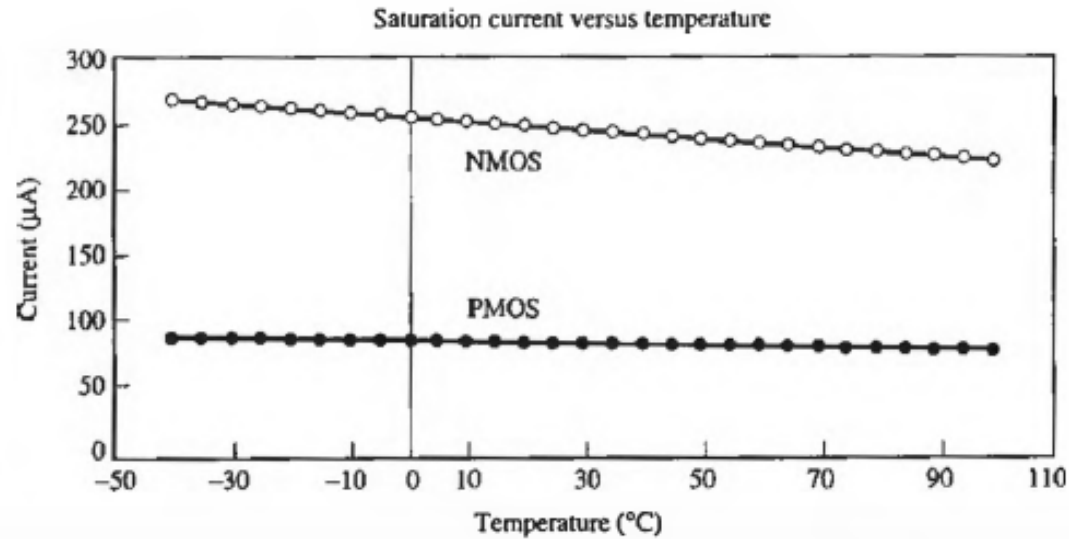
$$\mu(T) = \mu_0 \left(\frac{T}{300^\circ\text{K}} \right)^{-1.5}$$

- Intrinsic Carrier Concentration

This impacts the subthreshold current.

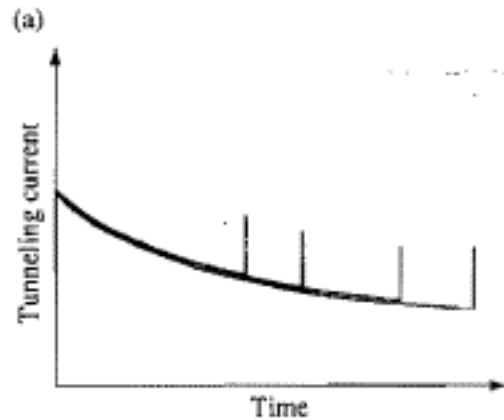
$$n_i = 1.45 \times 10^{10} \left(\frac{T}{300^\circ\text{K}} \right)^{1.5} e^{(1.12/0.0516 - E_g(T)/2V_{th})}$$

Temperature Effects



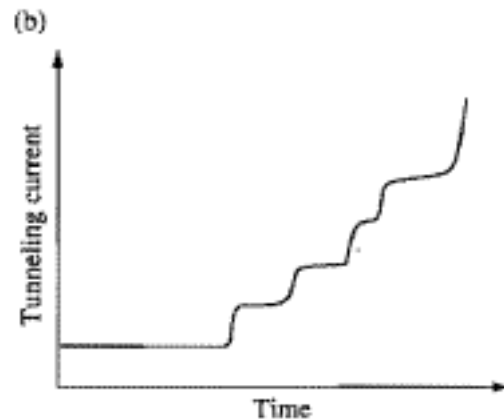
Thin vs. Thick Gate Oxide Breakdown

Thick Oxide



***Taur and Ning,
Fundamentals of VLSI
Devices, 2009**

Thin Oxide



Gate Oxide Breakdown

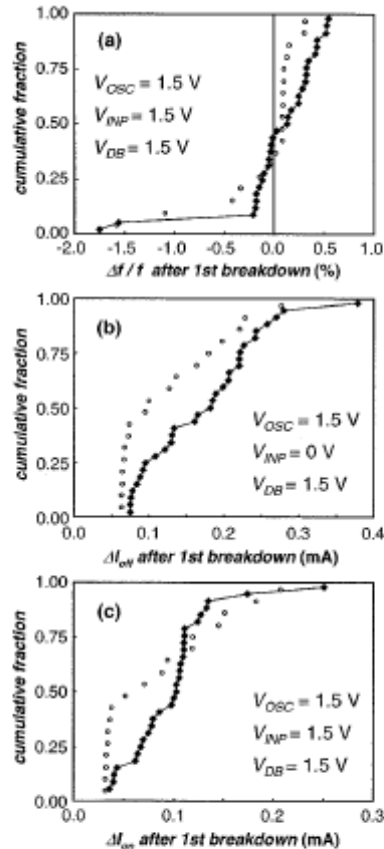


Fig. 5. Measured distributions of changes in (a) output frequency and (b) standby and (c) operating currents after the first hard gate oxide breakdown obtained on 31 oscillators (solid diamonds) are well matched by SPICE simulations of a ring oscillator with one "broken" nFET represented by an equivalent circuit for nFET after gate oxide breakdown (open circles).

***Kaczer, IEEE
TED, 2002**

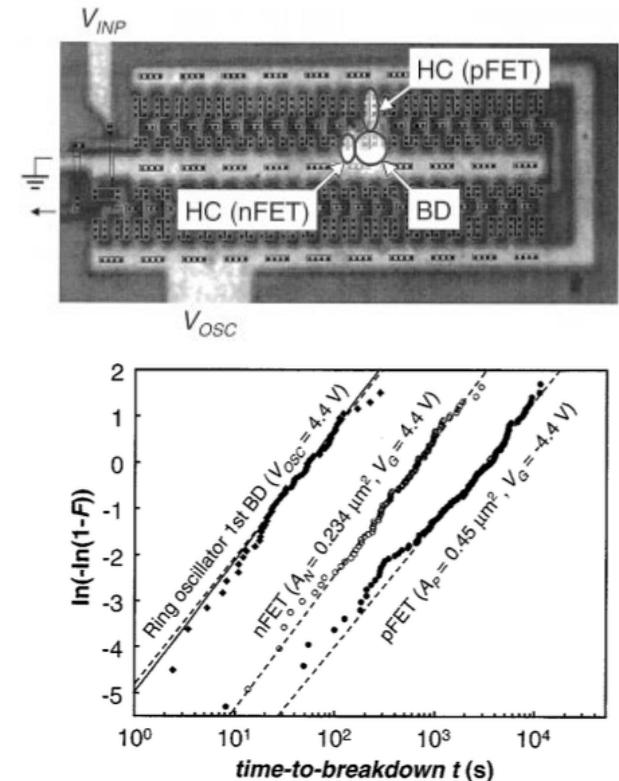
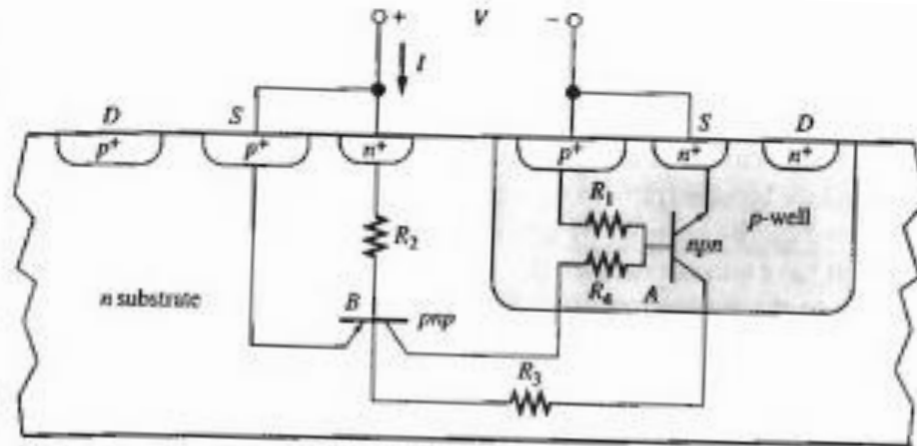
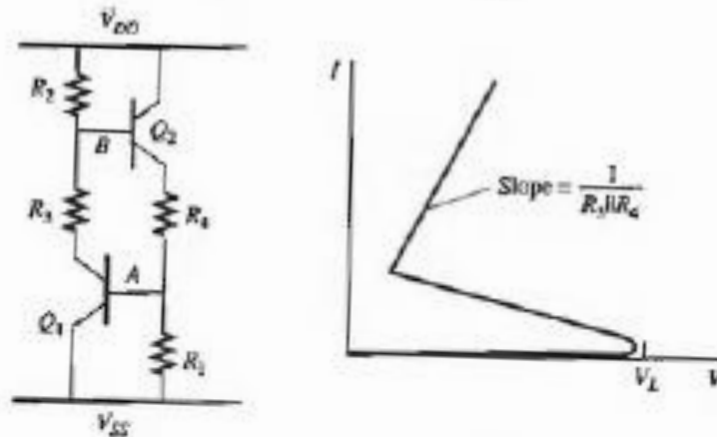


Fig. 6. Measured cumulative distribution of the time-to-first-breakdown of ring oscillator circuits (diamonds) is excellently matched by the distribution F_{OSC} (solid line), obtained by combining and scaling f_{BD} distributions of individual n- and pFETs under stress. Dashed lines represent maximum-likelihood Weibull fits to the respective data.

CMOS Latchup



(a) Origin of parasitic elements



(b) I - V characteristic of pnp - npn structure

Silicon-On-Insulator (SOI) Technology

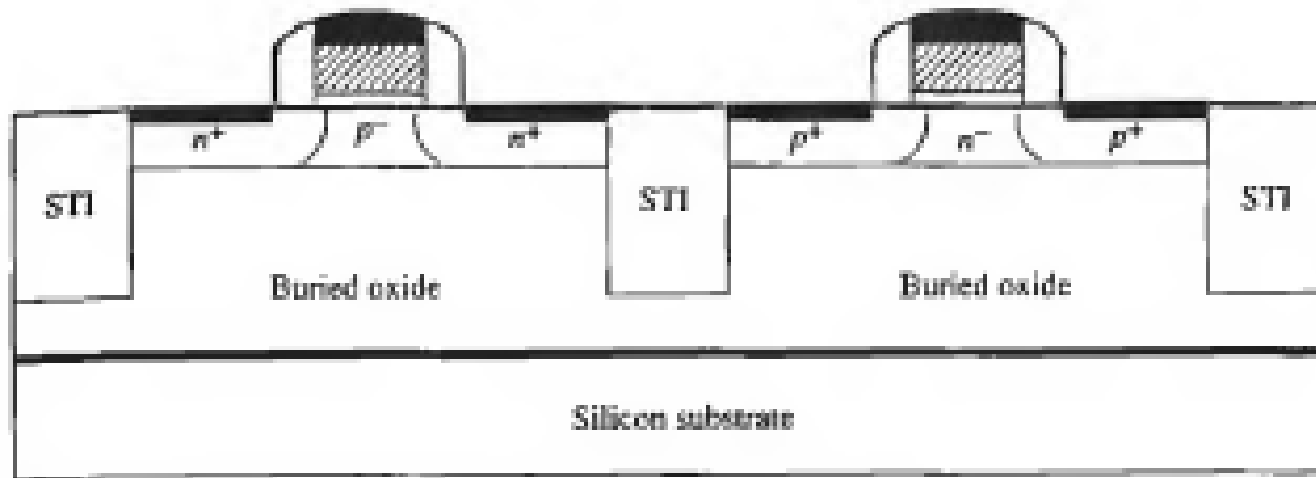


Figure 3.16

Structure of CMOS-SOI technology.

- Advance device performance by half a technology generation due to reduction in substrate capacitance
- More expensive substrate cost and poorer thermal heat sink